



THE UNIVERSITY OF
NOTRE DAME
A U S T R A L I A

MED 4000

WEEK 27

TUTOR GUIDE

The cases and materials in this guide were written and developed by;

Dr Carmella Law – Head Curriculum development and support.

The cases and materials in this guide were reviewed by;

- € Members of the University of Notre Dame clinical year's curriculum committee [CYC]: Last Reviewed February 2011.
- € Dr Ron Bova – ENT surgeon St Vincent's Hospital

**These resources are for the use of tutors and students,
University of Notre Dame Australia, School of Medicine, Sydney
Reproduction of the contents of these resources for any other
purpose is not permitted.**

RELEVANT PRIOR LEARNING:

MED1000:

- 1.31 History and examination of the ENT [L & CCS]
- 1.31 Anatomy & Physiology of hearing [L]
- 1.31 Bacterial and viral pathogens of the URT & Ear [L]

MED 2000

- 2.06 Head & Neck Anatomy [L]
- 2.06 Assessment of neck lumps [L]
- 2.30 Applied anatomy of the neck [L]
- 2.30 Imaging in thyroid disease [L]
- 2.30 Examination of the patient with thyroid disease [CCS]

MED 3000

- 3.17 Hypothyroidism [short case tutorial]
- 3.17 Hyperthyroidism [short case tutorial]
- 3.17 Neck swelling_goitre [short case tutorial]

WEEK 27_SURGERY_ENT		
Yr	Wk	LEARNING OBJECTIVES:
4	27	- List the commonest causes of sensorineural and conductive hearing loss in adults - Demonstrate an understanding of the applied anatomy of the hearing apparatus by discussing the broad categories of hearing aids that are available in Australia. Know the process by which patients can gain access to hearing aid devices. - Define tinnitus. Outline the initiators and aggravators of tinnitus. - List the 3 main features (associated with hearing loss) in a presentation that would require a referral to a specialist otolaryngologist.
4	27	- Know the infective and non-infective causes of otitis externa and the predisposing factors. - List the key differences between otitis externa and otitis media. - Discuss the two circumstances when syringing must <u>not</u> be carried out and why. - Outline the issues of concern when topical versus systemic antimicrobial therapy, for instance, in the treatment of otitis externa.
4	27	- List the key features of the history and examination in the assessment of a patient presenting with neck lumps. - Outline the differential diagnoses of neck lumps depending on the age of the patient. - Know the two investigations a general practitioner can conduct prior to referral to an otolaryngologist in the case of a patient presenting with neck lumps.
4	27	- Outline the differential diagnoses to be considered in a patient presenting with facial pain. - Briefly describe the symptoms which would steer the doctor towards the diagnosis of a dental abscess
4	27	- Know the features in history-taking that can assist in determining the differential diagnosis of a patient presenting with hoarseness. - Briefly outline the techniques used in voice clinics by otolaryngologists and speech pathologists in investigating patients with voice disorders.
4	27	- Know the commonest presentations of patients with thyroid nodules who are symptomatic. - Describe the key features of the medical history in a patient with thyroid nodules which would raise suspicion of malignancy. - Describe the two main investigations commonly used to assess thyroid nodules. - Know the indications for thyroidectomy in patients presenting with thyroid nodules. - Outline the broad categories of complications following thyroid surgery. - Be able to describe the key clinical features in a patient with a thyroid nodule which would increase suspicion of thyroid malignancy.

CASE ONE

Short case number: 4_27_01

Category: Respiratory and ENT systems

Discipline: Surgery_ENT

Setting: General practice

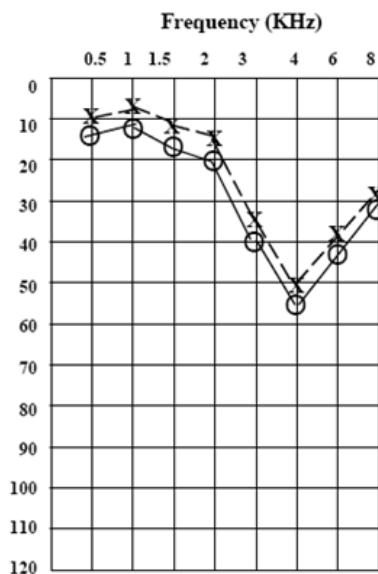
Topic: Assessment of hearing loss and tinnitus

CASE

Colin O'Farrell is a 50 year old worker in the construction industry who was persuaded by his wife to go to the surgery because his hearing had been deteriorating gradually over recent years. His hearing loss has become an issue in the past year because he has been turning up the television to a volume that the rest of the family found intolerable and they have also noticed that he has become more withdrawn of late, often avoiding social gatherings.

QUESTIONS

1. How prevalent is hearing impairment in Australia? List the commonest causes of adult hearing loss in sensorineural and conductive hearing loss categories. What are the main categories of underlying pathologies associated with sensorineural hearing loss (SNHL)?



Colin was referred to an audiologist for assessment of hearing loss, and the audiogram showed the characteristic appearance of associated with noise-induced hearing loss i.e. bilateral symmetrical downward sloping graphs with maximum loss at 4000 Hz.

When he was confronted with the diagnosis of a definite hearing loss, Colin was adamant against wearing hearing aids.

2. Describe the type of information would you give Colin. What types of hearing aids are available and what is the process for obtaining one in Australia?
3. What is tinnitus? What are the broad categories of causes and aggravators? How is this condition managed?
4. What features (associated with hearing loss) would alert you to the need to refer the patient to an otolaryngologist for further investigations?

Resources:

1. Kumar P & Clark M. Clinical Medicine, 7th edition. Chapter 20, The special senses p 1078-9. Saunders Elsevier 2009.
2. Macfarlane P and Carney AS. How to treat: Hearing loss in adults. Australian Doctor, August 2007, pp 29-36.
3. Tonkin J. Tinnitus. Australian Family Physician, 2002 Vol 31, No. 8.
4. Black B, Harvey L and Black J. How to treat: Devices to aid hearing. Australian Doctor, June 2004, pp 31-38.

ANSWERS

Question 1

How prevalent is hearing impairment in Australia? List the commonest causes of adult hearing loss in sensorineural and conductive hearing loss categories. What are the main categories of underlying pathologies associated with sensorineural hearing loss (SNHL)?

Hearing loss is the most prevalent sensory deficit reported by patients. 10-20% of Australians have some hearing impairment and approx 50% of those over 65 years of age are hearing impaired. Hearing loss is divided into 3 categories: sensorineural, conductive and mixed. Sensorineural hearing loss is further divided into sensory (cochlear) and neural (auditory nerve and pathway) causes.

The commonest causes of hearing loss in adults are:

A. Sensorineural

- **Noise-induced hearing loss**
- Presbycusis
- Sudden onset sensorineural hearing loss
- Acoustic neuroma

B. Conductive

- Chronic otitis media with effusion
- Cholesteatoma
- Tympanic membrane perforations
- Otosclerosis

The main categories of conductive and sensorineural deafness are:

Conductive	Sensorineural
Congenital – atresia Pendred's syndrome (see p. 985) Long QT syndrome (see p. 727) Björnstad syndrome (pili torti)	End organ Advancing age Occupational acoustic trauma Ménière's disease Drugs (e.g. gentamicin, furosemide)
External meatus Wax Foreign body Otitis externa Chronic suppurative	Eighth nerve lesions Acoustic neuroma Cranial trauma Inflammatory lesions: Tuberculous meningitis Sarcoidosis Neurosyphilis Carcinomatous meningitis
Drum Perforation/trauma	
Middle ear Otosclerosis Ossicular bone problems Suppuration (otitis media)	Brainstem lesions (rare) Multiple sclerosis Infarction

Question 2

Describe the type of information would you give Colin. What types of hearing aids are available and what is the process for obtaining one in Australia?

NOISE-induced hearing loss due to exposure at work, recreation or in the home is one of the most common causes of hearing impairment worldwide. Although not all people exposed to loud noise develop noise-induced hearing loss, there is a strong correlation between prolonged exposure to noises >85dB and hearing loss. People employed in industrial workplaces and wearing no hearing protection are exposed to noises in the 90- 100+ dB range, with some hand-operated machinery producing noise up to 120dB.

The effect of such daily exposure over several years is cumulative and can result in permanent damage to the auditory nerve. It can be avoided by use of well-fitting earmuffs or earplugs, which reduce the sound intensity levels by 15-25dB on average, usually bringing the noise thresholds back to an acceptable 85dB. As a rule, for every 3dB increase over 85dB, the maximum acceptable time to be exposed to the noise decreases by half. For example, in an 88dB environment a worker has a maximum exposure time of 4 hours, and two hours in a 91dB environment.

Adequate, well-fitting hearing protection is essential for preventing noise-induced hearing loss and should be worn both in the workplace and during activities or hobbies that involve further exposure to loud noises. Avoiding further exposure to **medications known to be ototoxic (Table below)** is also important in patients with hearing loss.

Drug class or chemical agent	Example	Comment
Aminoglycoside antibiotics	Gentamicin, tobramycin, amikacin	More likely with prolonged course or high doses and in people with renal impairment. May also cause vestibular damage
Loop diuretics	Frusemide	Reversible dose-related hearing loss. More likely in people with renal impairment
Alkylating agents	Cisplatin	Hyperpolarisation of hair-cell membranes and raised auditory thresholds
Salicylates	Aspirin	Reversible sensorineural hearing loss
Others	Phenytoin, beta blockers, quinine	Documented ototoxic potential
Heavy metals	Arsenic, cobalt, lead, lithium, mercury, thorium	Documented ototoxic potential
Chemicals	Cyanide, benzene, aniline dyes, iodine, carbon tetrachloride	May be ototoxic
Solvents	Toluene, styrene	May be ototoxic

Hearing aids may be of benefit in correcting the noise-induced hearing loss due to occupational exposure at work in Colin's case. For many patients the fitting of an appropriately configured hearing aid results in a significant improvement in hearing. Patients with a moderate to severe hearing impairment subjectively benefit more from hearing aids than those with only moderate impairment. The audiologist plays an important role in choosing the most appropriate type of hearing aid for the needs of a given patient. In general, patients with low-tone hearing loss receive the greatest quality-of-life benefit from hearing aids. The circumstances of each individual patient need to be assessed before a decision on a hearing aid trial.

Hearing aids (cochlear hair-cell stimulators)

Air conduction

- Extracorporeal
- In-the-ear aids
- Ear level or body worn
- Behind-the-ear aids

Bone conduction

- Headband models
- Bone-anchored hearing aids
- Semi-implantable models (Xomed audient)

Assistive/alertive

- Telephone attachments
- Sound
- Video displays
- FM loops
- TV subheadings
- Vibro-tactile devices
- Non-acoustic aids, eg, flashing-light doorbells

Semi-implantable ossicle vibrators

- Free attachments
- Drum lens
- Ossicle attachments
- Connected vibrators
- Electromagnetic/piezoelectric devices
- RetroX (small behind-the-ear pack connected to the lumen of the external canal via a transcutaneous titanium cannula)

Electronic/bionic implants

Cochlear implants

- Extracochlear
- Intracochlear (single or multiple electrodes)

Brainstem implants

Hearing aids are subdivided into air- and bone-conduction variants. The most common and clinically available hearing aids are air-conduction aids. In these devices a signal is amplified by an ear-level or body-worn device, which produces amplified sound that passes through the drum and ossicles via the external auditory canal.

These can be used for both sensorineural and conductive losses. Bone conduction aids (worn as either a headband or as an implantable device) deliver an amplified signal via bone conduction of sound through the mastoid to the cochlea. **After medical and audiological evaluation, a hearing aid can be obtained by one of several routes.** In Australia, pensioners, gold-card and entitled white-card veterans, those receiving Centrelink sickness allowance benefits, and the dependants of these groups, may obtain aids free of charge. Military personnel are also eligible.

Application forms (2647–0008) are available from the Commonwealth Office of Hearing Services (www.health.gov.au/hear; Ph: 1800 686126). The OHS assesses the application for eligibility, and appropriate patients receive a voucher to obtain an aid, either from an Australian Hearing Services branch or from approved firms (listed with the voucher documents).

Question 3

What is tinnitus? What are the broad categories of causes and aggravators? How is this condition managed? Tinnitus is the hearing of sound (or noise) generated from within the head.

Tinnitus may be objective in which case an observer (the physician) can also hear the sound, especially with a stethoscope. The more common non-objective tinnitus is only heard by the patient. It is mainly nonpulsatile. Abnormally elevated spontaneous discharge rates in single neurons are often observed in cochlear (sensori-neural) noise damage, when damage is confined to the outer hair cells leading to aberrant discharge levels.

The cause of abnormal signals are called the **initiators** and may originate from: injury, infection, tumours, degeneration, Meniere's disease.

Aggravators of tinnitus include:

- temporomandibular joint disorder (TMJ)
- malocclusion
- injury
- arthritis
- middle ear disease
 - serous otitis media
 - Eustachian tube dysfunction

- tympanosclerosis
- allergic rhinitis and sinusitis
- stress
- depression.

Often, there is no definitive cause found for tinnitus and the aim of management becomes the minimisation and control of symptoms rather than cure. In these cases the management of tinnitus should be holistic. Generally no one approach is sufficient. It is important at this stage to reassure the patient that the tinnitus is nearly always amenable to treatment. The mainstay of treatment is masking techniques. Patients should avoid silence so that their brain processes ambient sounds rather than focusing on tinnitus. Having music on in the background is the most practical masking technique. Eventually patients train their own brain to ignore tinnitus.

- Mineral supplements such as zinc and magnesium are frequently used in the medical treatment of tinnitus, although there is no evidence to support its use.
- In patients with more severe tinnitus clonazepam 0.5 mg given in the evenings can be helpful. However counselling should be tried first.
- Additional management of tinnitus by means of a hearing aid in the affected ear is a potent option. Hearing aids are useful if patients have significant hearing loss associated with tinnitus.
- Occasionally treatment by a clinical psychologist is necessary in the management of tinnitus. This may involve tinnitus retraining therapy, cognitive behaviour therapy and even hypnosis.

Question 4

What features (associated with hearing loss) would alert you to the need for referral to an otolaryngologist for further investigations?

Any of the following features in the history and/or on physical examination of the patient with hearing loss necessitate referral to an otolaryngologist for further investigation:

1. asymmetric presentation

- unilateral or asymmetrical hearing loss
- unilateral or pulsatile tinnitus

2. chronic or persistent symptoms

- chronically discharging ear
- persistent middle-ear effusion
- non-healing tympanic membrane perforation

3. suggestion of more serious underlying pathology

- suspicion of cholesteatoma
- postural imbalance or vertigo
- large exostosis
- conductive hearing loss and normal physical examination (for otosclerosis)
- facial weakness

CASE TWO

Short case number: 4_27_02

Category: Respiratory and ENT systems

Discipline: Surgery_ENT

Setting: General practice

Topic: The discharging ear_otitis externa

CASE

Susan is a 45 year old bank teller who has put on some weight and decided to get fit by joining the swimming classes at the local gym over the past month. She presents to you because she has noticed that her left ear was oozing for a couple of days. On further questioning, she said she had noticed that initially her ears felt wet, so she tried to clean them with cotton tip applicators.



This made the situation worse and now the left ear is sore and she can't hear very well.

On examination, Susan was afebrile and you confirmed the discharge from the left ear. You also noticed that there was some pain when you gently pull the pinna of the left ear. There were also some tender preauricular nodes. You couldn't examine the left ear properly because of the pain but you noted a lot of wax in the right ear which was not tender.

QUESTIONS

1. What are the infective and non-infective causes of discharge from the external ear? What are the predisposing factors for otitis externa? Discuss the differential diagnoses for otitis externa.
2. The impacted wax in the infected ear prevents proper otoscopic examination. Why should you not syringe the ear in this situation?
3. Compare and contrast the flora of the normal ear with that of the infected ear. What is the current thinking when considering topical versus systemic antimicrobial therapy in the treatment of acute otitis externa?

Resources:

1. Kumar P & Clark M. Clinical Medicine, 7th edition. Chapter 20, The special senses p 1077-8. Saunders Elsevier 2009.
2. Taplin MA. How to treat: External ear conditions. Australian Doctor, April 2007, pp 25-32.

ANSWERS

1. What are the infective and non-infective causes of discharge from the external ear? What are the predisposing factors for otitis externa? Discuss the differential diagnoses for otitis externa.

Table 3: Classification of otitis externa			Table 4: Predisposing factors for otitis externa	
Infective	Bacterial	Acute diffuse	Predisposing factor	Examples
		Acute localised (furunculosis)	Water exposure	Seasonal humidity ('tropical ear'), swimming, water sports
		Chronic	Wax	Impaction, removal or loss of physical protection, change of pH
	Fungal	Otomycosis	Dermatological	Eczema, seborrhoeic, psoriasis
	Viral	Herpes zoster oticus	Trauma	Foreign bodies, cotton buds, paper clips, safety pins
Non-infective (inflammatory)	Dermatitis	'Malignant'	Anatomical	Narrow/long/hairy ear canals, stenosis, exostoses
		Seborrhoeic	Localised ear disease	Tympanic membrane perforation discharge, cholesteatoma
		Atopic-eczematous, iatrogenic	Systemic disease	Diabetes, immunocompromised
		Psoriasis	Radiotherapy	Head and neck, CNS malignancy
			Assistive hearing devices	Hearing aids, ear plugs, diving suits, mobile phone ear pieces

Discharge from the ear is usually due to infection of the outer or middle ear. Otitis externa is a diffuse inflammation of the skin of the ear canal. The organism may be bacterial, viral or fungal and the patient usually complains of severe pain. Treatment is with regular cleansing and topical antibiotics combined with corticosteroids; it resolves in 3–4 days.

Table 7: Differential diagnosis of acute otitis externa

Clinical presentation	Otitis externa	Acute otitis media (with perforation)	Chronic suppurative otitis media*
History			
Pain	Often severe	Before perforation	Uncommon, unless complication present
Fever	Rare	Before rupture	Rare
Discharge	Thick, mucopurulent	Often copious, may be serosanguineous	Purulent, chronic or recurrent
Associated features	Swimming, warm or humid weather	URTI, grommet in situ	Chronic tympanic membrane perforation ± cholesteatoma
Season	Summer	Winter or spring	Variable
Physical examination			
Ear canal	Tender tragus or pinna, oedema, erythema	Copious discharge. Pain is related to associated secondary otitis externa	Chronic discharging ear, offensive odour. Pain might indicate complication
Tympanic membrane	Normal or mild inflammation. May be difficult to visualise	Perforated, red, inflamed, opaque. Gentle forced Valsalva manoeuvre may produce bubbles	Larger perforation, squamous or keratin debris. Consider cholesteatoma
Organisms	<i>Pseudomonas aeruginosa</i> , <i>Staphylococcus aureus</i> (skin pathogens)	<i>Streptococcus pneumoniae</i> , <i>Haemophilus influenzae</i> , <i>Moraxhella catarrhalis</i> (respiratory tract pathogens)	<i>Pseudomonas aeruginosa</i> , <i>Staphylococcus aureus</i> , coliforms, others

*Includes tympanostomy tube otorrhoea

2. The impacted wax in the infected ear prevents proper otoscopic examination. Why should you not syringe the ear in this situation?

Syringing is contraindicated when there is gross wax impaction in the deep ear canal or when tympanic membrane perforation cannot be excluded. It should also be avoided in patients with external ear canal infections, dermatitis of the ear canal, previous ear surgery and in patients with stenotic ear canals. Complications of ear syringing include perforation of the tympanic membrane, laceration to the ear canal skin and secondary otitis externa.

Ear syringing should only be performed by experienced doctors. Ideally the wax is initially loosened with the aid of a cerumenolytic. ENT surgeons will often remove wax and keratin with the aid of a microscope. Wax curettes and gentle suction ear toilet is generally well tolerated and safe. Patients with tight stenotic ear canals or other ear disease such as tympanic membrane perforation should be reviewed by a specialist for aural toilet and further management.

3. Compare and contrast the flora of the normal ear with that of the infected ear. What is the current thinking when considering topical versus systemic antimicrobial therapy in the treatment of acute otitis externa?

Table 1: Microbiology of the healthy and infected ear

Micro-organism	Healthy ear (% of normal flora)	Otitis externa (% of normal flora)
<i>Staphylococcus epidermidis</i>	33-100%	<5%
<i>Corynebacterium/diphtheroids</i>	<33%	<5%
Alpha-haemolytic streptococci	<33%	<5%
<i>Staphylococcus aureus</i>	<33%	<33%
Anaerobes (<i>Escherichia coli</i> , <i>Proteus</i>)	<33%	Unknown
<i>Klebsiella</i> , <i>Enterobacter</i>	<5%	<33%
<i>Pseudomonas aeruginosa</i>	<5%	33-100% (more common in swimmers)
<i>Aspergillus</i> , <i>Candida</i>	<5%	0-66% (more common in tropical climates)

The normal flora of the ear canal is similar to that of the skin. Water exposure and excessive cleaning of the canal will alter the microbiological balance of the external ear. *Staphylococcus epidermidis* is the most common commensal organism. *Pseudomonas aeruginosa* and *Staphylococcus aureus* are the most common pathogens found in infected ears.

In 2006, the American Academy of Otolaryngology – Head and Neck Surgery produced an evidence-based guideline* for the treatment of acute otitis externa, based on numerous clinical and scientific studies as well as large meta-analysis. The information below is adapted from this guideline.

There is level B evidence supporting the use of topical versus systemic therapy in the initial stages. **Topical preparations are recommended for uncomplicated acute otitis externa** (no osteitis, abscess, middle-ear disease, tympanic membrane perforation or recurrent episodes of infection). **Coexistent diabetes, immunosuppression or spread of disease beyond the ear canal into the pinna, skin of the face or neck or deeper tissues (malignant otitis externa) requires referral to an ENT specialist and use of oral or IV therapy.**

Studies have shown that 20-40% of patients with acute otitis externa receive oral antibiotics, often as well as topical antimicrobials. Many of the oral antibiotics are inactive against *Pseudomonas* and *S aureus*, the most common pathogens. Treatment with penicillins, cephalosporins or macrolides increases disease persistence, and treatment with cephalosporins also increases recurrence.

Otological therapies have several advantages over systemic therapy

for uncomplicated, diffuse acute otitis externa. These include:

- High concentration of medication at the site of infection.
- Low risk of developing antibiotic resistance.
- Minimal systemic adverse effects.
- Relative ease of administration.

Level B evidence supports the following recommendations regarding choice of topical therapy.

It is not always necessary to treat diffuse acute otitis externa with an antibiotic. Several acidifying solutions are available, including acetic acid and gentian violet.

Topical agents are inexpensive and have been used for many years, particularly in developing countries. They are unlikely to cause bacterial resistance. Otological antibiotics contain either an aminoglycoside (gentamicin, neomycin, tobramycin, polymyxin B and framycetin) or fluoroquinolone (ciprofloxacin, ofloxacin). Aminoglycosides have been around for longer, are generally cheap and have a good antibacterial spectrum for acute otitis externa but in recent years; aminoglycoside ototoxicity (cochlear and vestibular) has become more apparent. Aminoglycosides and topical alcohol-containing agents should not be used when the tympanic membrane is perforated and the middle-ear space is open. The risk of ototoxic injury outweighs the benefits compared with non-ototoxic antimicrobials with equal efficacy.

With otological antimicrobial treatments for acute otitis externa, **meta-analysis has shown no significant difference in clinical outcomes (clinical and bacteriological cure rates) for patients with acute otitis externa with regard to use of:**

- Antiseptic versus antimicrobial.
- Quinolone vs. non-quinolone antibiotic.
- Steroid-antimicrobial vs. antimicrobial alone.

Regardless of the topical agent used, 65-90% of patients have clinical resolution within 7-10 days. Recent studies have shown significant differences in the rapidity of treatment response and symptom resolution. The addition of hydrocortisone to ciprofloxacin significantly reduced median ear pain by one day.

CASE THREE

Short case number: 4_27_03

Category: Respiratory and ENT systems

Discipline: Surgery_ENT

Setting: General practice

Topic: Neck lumps (refer also to short case 4.20.02)

CASE

Peter is a 45 year old bank teller who presents with a 3 months history of an enlarging mass in the upper neck on the right side, which is painless. His history includes smoking 20 cigarettes a day for the past 20 years.



Examination confirms a 4cm mass in the neck which has a firm consistency. Oral examination reveals leukoplakia of the tongue but no areas of hardening or ulceration. There were no palpable lymph nodes.

QUESTIONS

1. List the key features which should be covered in the history and examination of a patient presenting with neck lumps.
2. Revise the anatomy of the neck and outline the causes of neck lumps depending on the age of the patient.
3. Discuss the two investigations a GP can arrange prior to referral.

Resources:

1. Hobbs C and Bova R. Neck lumps. *Medicine Today*, April 2010, 11(4): 26-34.
2. Hughes C and O'Brien C. How to treat: Neck lumps. *Australian Doctor*, August 2005, pp 31-38.

ANSWERS

1. List the key features which should be covered in the history and examination of a patient presenting with neck lumps.

Hobbs & Bova article: In the assessment of a patient presenting with neck lumps

- Take a full history including duration, change in size of the lump, dysphagia, hoarseness, referred pain, fever, night sweats and weight loss. Check risk factors including smoking, alcohol consumption, upper aerodigestive tract malignancy and sun exposure.
- Examine the seated patient – inspect the neck from the front and palpate from behind. Is the lump a midline or lateral neck lump? Is it soft or hard? Is it mobile or fixed? Complete the H&N examination – remainder of the neck, thyroid, oral cavity, oropharynx, nose and ears, skin of the scalp, back of the neck. Examine the facial nerve for lumps in the parotid region.

While examining the tongue, oral cavity and oropharynx it is important to look for:

- Any mucosal changes associated with smoking, such as dysplasia or leukoplakia.
- Any discrete masses or lesions, such as a mass in the tongue or floor of the mouth.

2. Revise the anatomy of the neck and outline the causes of neck lumps depending on the age of the patient.

Causes of neck lumps, according to age				
	Infant	Child	Adolescent	Adult
Neoplastic		Neoplastic (parotid, thyroid)	Lymphoma (Hodgkin's)	Metastatic cancer, lymphoma (Hodgkin's/ non-Hodgkin's), thyroid
Infective/inflammatory	Non-specific lymphadenitis	Non-specific lymphadenitis, cat scratch disease, toxoplasmosis, atypical mycobacteria, mumps (viral parotitis)	Glandular fever, dental infection, cat scratch disease, toxoplasmosis, mumps (viral parotitis)	Dental infection, HIV lymphadenopathy, toxoplasmosis, cat scratch disease, TB
Congenital	Cystic hygroma, vascular malformation, congenital torticollis	Dermoid cysts, vascular malformations	Thyroglossal cyst, branchial cysts	Congenital sebaceous cyst

Note that any adult with a lump in the lateral neck has cancer until it is proved otherwise.

Figure 2: Triangles, lymph node levels and normal lymph nodes in the neck.

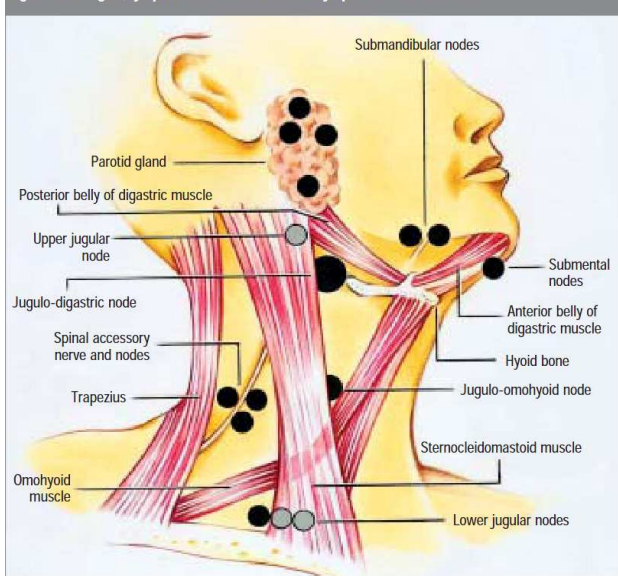
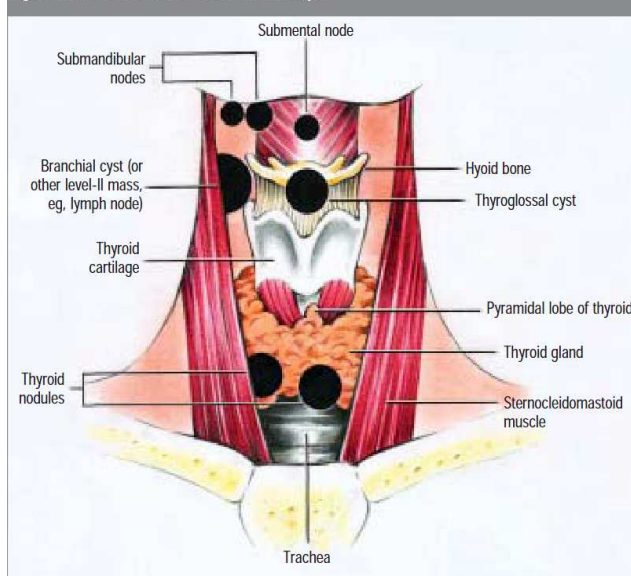


Figure 3: Anatomical sites of anterior neck lumps.



3. Discuss the two investigations a GP can arrange prior to referral.

The 2 most important investigations that a GP can arrange prior to referral of a patient with a neck lump are neck CT scan and a fine needle aspiration biopsy.

a) Soft tissue CT scan with contrast is very helpful in investigating neck lumps because it provides superior anatomical definition of any neck lump while also imaging the remainder of the neck tissues. If there are any metastatic nodes, this procedure may also facilitate localisation of the primary tumour.

b) Fine needle aspiration biopsy (FNAB) is the most accurate diagnostic tool for investigating neck lumps. It can be performed in most radiology centres (under ultrasound guidance), in major pathology centres or by surgeons in their rooms. Although the accuracy of FNAB is high (approx 90%), false negatives do occur and hence **a suspicious neck mass should always be referred for comprehensive evaluation**

Once referred to the ENT specialist, the following investigations may be carried out:

Incision and excision biopsy of neck lumps should not be carried out in a non-specialist environment. If the lump proves to be malignant, inappropriate biopsy could lead to difficulty in later treatment and increased risk of recurrence.

Magnetic resonance imaging (MRI) is not widely used for assessment of neck lumps although this procedure has particular strength when accurate information on soft tissue pathology is required. The use of magnetic resonance angiography is also helpful in assessing vascular malformations in the head and neck.

Positron emission tomography PET¹ imaging, or PET scan is particularly valuable in investigating metastatic disease with unknown primary source and for detecting recurrent tumour. It should not be used in general practice. In specialist hands it is useful for staging head and neck cancers and also to pick up metastases elsewhere before commencement of therapy.

¹ PET is a diagnostic examination that involves the acquisition of physiological images based on the detection of positrons emitted from a radioactive substance administered to the patient (18FDG [fluorodeoxyglucose]).

CASE FOUR

Short case number: 4_27_04

Category: Respiratory and ENT systems

Discipline: Surgery

Setting: General practice

Topic: Facial pain_ dental abscess [SDL]

CASE

Christopher is a 40 year old office manager who presents with pain over the right cheek. It started as a mild ache the previous evening and he thought it was due to the temporo-mandibular joint problem, which was diagnosed a few years before. However that morning the pain seems to have moved away from the ear towards the cheek. On examination, you noted tenderness over the right cheek. The gum around one of the molars appeared inflamed and there was severe pain when you tapped on the tooth.



QUESTIONS

1. What are the differential diagnoses that need to be considered in a patient presenting with facial pain?
2. Outline the symptoms would steer you in the direction of a dental diagnosis, particularly a dental abscess.

Resources:

1. Quail G. Atypical facial pain. *Australian Family Physician*, 2005 34(8): 641-5.

ANSWERS

1. What are the differential diagnoses that need to be considered in a patient presenting with facial pain?

Facial pain is frequently misdiagnosed, subjecting the patient to prolonged and costly treatment. Undoubtedly dental and oral diseases including temporo-mandibular joint (TMJ) dysfunction are the commonest causes, and atypical facial pain should be regarded as dental in origin until proven otherwise. However, there are many other causes as shown in Table 1 below.

Table 1. Causes of facial pain	
Dental, oral, pharyngeal	Infective causes – dental caries, periapical infection – periodontal disease, mucosal disease – herpes simplex, cellulitis – tonsillitis Traumatic – fractured tooth Neoplastic – primary carcinoma, metastatic disease Inflammatory lesions – aphthous ulcer, lichen planus
Temporo-mandibular joint dysfunction	
Facial bone disease	Infective – osteomyelitis Traumatic – fractures
Salivary gland	Infective – parotitis Metabolic – calculus formation – neoplastic
Paranasal sinus disease	
Neurological disorders	– including trigeminal and glossopharyngeal neuralgias – compression neuropathies – postinfective conditions: herpes zoster – migraine
Vascular	– cranial arteritis – angina pectoris
Psychogenic	

Pain in and around the eye, when not caused by ocular disease, should be considered as a headache. Rarely, inflammatory or infiltrative lesions at the apex of the orbit or the cavernous sinus may cause pain in or around the eye. Pain in the eye may accompany disorders of the carotid artery, particularly dissections, and may then be accompanied by a Horner's syndrome.

Pain in other parts of the face can be due to problems with the teeth or the **temporo-mandibular joint (TMJ)**². Inflamed nasal sinuses are seldom the cause of lasting facial pain in the absence of obvious nasal congestion. The very rare but serious condition of subdural empyema needs to be considered if 'sinusitis' is followed by very severe unilateral facial pain and signs of cerebral irritation (seizures and/or obtundation). Destructive lesions of the trigeminal nerve causing pain are extremely rare.

Most patients with persisting pain in the face have trigeminal neuralgia, atypical facial pain or post-herpetic neuralgia. The main distinction between these is in the nature of the pain. In trigeminal neuralgia, the pain is very brief, though severe and recurrent, described as 'like lightning' and is most frequently felt in the second and third divisions of the nerve. Atypical facial pain, on the other hand, is continuous and unremitting, and is centred over the maxilla, usually on the left side. It occurs most frequently in middle-aged women. Post-herpetic neuralgia is continuous and is felt as a burning pain throughout the affected territory, which is often very sensitive to light touch.

2. Outline the symptoms would steer you in the direction of a dental diagnosis, particularly a dental abscess. What is the appropriate treatment by a GP?

Symptoms	Likely cause	Initial management suggestions	Ongoing care
Tender to pressure	Possible early abscess formation	Antibiotics may be indicated	Root canal therapy or extraction
Tender/painful swelling in region of recent toothache	Dental abscess	- Analgesics - Antibiotics probably indicated - If swelling is causing dysphagia/dyspnoea, admit to hospital for aggressive intravenous antibiotic therapy and appropriate supportive management	- See a dentist urgently - Root canal therapy or extraction - Monitor closely – can be life threatening if cellulitis develops¹
Pain worsens when head proclined	Possibly originating from maxillary sinus (pus/exudate moves in sinus)	Antibiotics, inhalations and nose drops generally effective for acute maxillary sinusitis	
Pain persists 1–4 days after dental extraction	Alveolar osteitis ² (also known as 'dry socket')	- Antibiotics not indicated - Flush socket with sterile saline - Insert sedative dressing if available	- See a dentist urgently - Socket irrigation and sedative dressing

Dental abscesses, which should be primarily managed by drainage of pus, are of two types:

- periodontal (where antibiotics may be indicated)
- endodontic (where antibiotics are inappropriate as they will not reach a necrotic pulp).

Accurate diagnosis may be difficult and this can be determined by a radiology referral for an OPG. If antibiotics are indicated for the cellulitis, the patient may be commenced on broad spectrum antibiotics such as cephalosporin/penicillin or clindamycin if allergic to penicillin.

² This is a disorder of the TM joint associated with nocturnal tooth grinding or abnormalities of bite. It particularly occurs in anxious people. It gives rise to pain in one or both temporomandibular joints. Dental correction of the bite helps a few. Many patients are exposed to much unnecessary dental treatment.

CASE FIVE

Short case number: 4_27_05

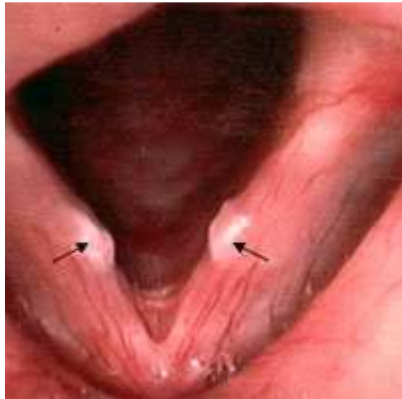
Category: Respiratory and ENT systems

Discipline: Surgery_ENT

Setting: ENT clinic

Topic: Assessing the patient with hoarseness

CASE



Sandra is a 35 year old waitress who enjoys karaoke with her friends on her nights off. She is a social smoker and most of her friends smoke. She presented to her GP recently because over the past 6 months she has noticed fluctuating hoarseness, which is worse after a night out. The rest of her history and her examination was otherwise normal.

Her GP referred her to the ENT clinic for laryngoscopy. Bilateral vocal cord nodules were detected.

QUESTIONS

1. What features in history-taking can assist you in working through the differential diagnoses of a patient presenting with hoarseness?
2. Name some of the newer techniques used in voice clinics by otolaryngologists and speech pathologists in investigating patients presenting with voice disorders.
3. Outline the management of this patient.

Resources:

1. Kumar P & Clark M. Clinical Medicine, 7th edition. Chapter 20. The special senses p 1077-8. Saunders Elsevier 2009.
2. Innes JA and Reud PT. Respiratory Disease Chapter 19, p 730-1 in Davidson's Principles and Practice of Medicine 20th Edition. Churchill Livingstone Elsevier 2006.
3. Bova R. Hoarseness a guide to voice disorders. Medicine Today. Feb 2007. Vol 8 (2): 38-44.
4. Havas TE. How to treat: Hoarseness and voice dysfunction. Australian Doctor. May 2006, pp 29-36.

1. What features in history-taking can assist you in working through the differential diagnoses of a patient presenting with hoarseness?

Dysphonia (hoarseness) refers to altered voice due to a laryngeal disorder and may be described as raspy, gravelly or breathy. The history of the change in voice provides clues to the cause.

Intermittent vs. constant/progressive: Intermittent dysphonia is normally always secondary to a benign disorder but constant or progressive dysphonia should always alert the doctor to the possibility of a malignancy.

Duration: hoarseness after a viral URTI causing laryngitis is likely to last only a few days, whereas huskiness caused by Reinke's oedema may last months or years.

Onset: the onset of hoarseness may be sudden, for example, when shouting causes haemorrhage of the vocal cord or when there is damage to the recurrent laryngeal nerve during thyroid surgery.

Progression: the progression of symptoms over weeks or months should cause particular concern and suggest the possibility of neoplasm.

Exposure to irritants: It is important to ask about exposure to any potential environmental irritants such as cigarette smoke, or excessive periods in air conditioning or cool rooms.

Variability: variable, irregular fluctuating voice production associated with loss of intelligibility may be the first manifestation of a neurological disorder.

There are 3 essential components for voice production: an air source (the lungs); a vibratory source (the vocal cords); and a resonating chamber (the pharynx, the nasal and oral cavities). Although chest and nasal disorders can affect the voice, **the majority of hoarseness is due to laryngeal pathology.**

Nodules (always bilateral and commoner in females) and polyps are found on the free edge of the vocal cord preventing full closure and giving a 'breathy, harsh' voice. They are commonly found in professions that rely on their voice for their livelihood, such as teachers, singers and lawyers. They are usually related to poor technique of voice production and can usually be cured with speech therapy. If surgery is needed, great care must be taken to remain in the superficial layers of the vocal cord in order to prevent deep scarring which may leave the voice permanently hoarse.

Reinke's oedema is due to a collection of tissue fluid in the subepithelial layer of the vocal cord. The vocal cord has poor lymphatic drainage, predisposing it to oedema. Reinke's oedema is associated with irritation of the vocal cords: smoking, voice abuse, acid reflux and very rarely hypothyroidism. Treatment is to remove the irritation in most cases but surgery to thin the cords will also allow the voice to return to its normal pitch.

Acute-onset hoarseness

Hoarseness, in a smoker, is a danger sign. Any patient with a hoarse voice for over 6 weeks should be seen by an ENT surgeon. The voice may be deep, harsh and breathy indicating a mass on the vocal cord or it can be weak suggesting a paralysed left vocal cord secondary to mediastinal disease, e.g. bronchial carcinoma. Early squamous cell carcinoma of the larynx has a good prognosis. Treatment is with carbon dioxide laser resection or radiotherapy. Spread of the tumour can lead to referred otalgia which may then require a laryngectomy with possible neck dissection. A patient with a paralysed left vocal cord must have a CT of the neck and chest. Medialization of

the paralysed cord to allow contact with the opposite cord can return the voice and give a competent larynx. This can be done under local anaesthesia, giving an immediate result whatever the long-term prognosis of the chest pathology.

The most common neurological disorders affecting voice are:

- Vocal cord paralysis
- Multiple sclerosis
- Motor neurone disease
- Myasthenia gravis
- Spasmodic dysphonia
- Parkinson's disease.
- Cerebellar degeneration
- Pseudo bulbar palsy
- Various forms of dystonias

The clinician must also be mindful of the possibility that laryngeal symptoms can be the initial presenting feature of generalised systemic diseases. Hoarseness can be caused by: Rheumatoid arthritis of the larynx affecting the cricoarytenoid joints; Sarcoidosis; Amyloidosis; Wagner's granulomatosis; Epidermolysis bullosa; Bullous pemphigoid.

2. Name some of the newer techniques used in voice clinics by otolaryngologists and speech pathologists in investigating patients presenting with voice disorders.

When symptoms suggest possible pathology in the larynx or pharynx, the patient should be referred to an otolaryngologist, who can:

- Examine the upper airway, including the nose and nasal cavity, mouth, oral cavity, pharynx and larynx.
 - Perform **indirect laryngoscopy** using a mirror.
 - Perform a **direct laryngoscopy** using a rigid telescope and flexible fiberoptic naso-laryngoscopy.
- In most cases, when anatomical abnormalities are found at the initial examination, a diagnosis can be made on the basis of the physical examination. However, patients with more complex functional voice disorders often need referral to a voice clinic.

Patients at most **voice clinics** are seen conjointly by an otolaryngologist and a speech pathologist, who take a detailed history of general health and specific voice problems. Perceptual analysis and complex computer acoustic analysis are used to break down the voice into specific acoustic parameters, which can then be studied individually. This type of acoustic diagnosis may be useful in assessing some conditions such as vocal nodules, when there is a characteristic pattern of dysphonia called diplophonia, in which the vocal cords vibrate at two discrete fundamental frequencies.

During acoustic diagnosis patients are asked to phonate and/or sing a series of pre-arranged texts that emphasise every aspect of laryngeal muscle use. The larynx is examined in detail using rigid and flexible telescopes and the images are projected onto a computer screen and recorded to facilitate detailed computer analysis. During *video endostroboscopy* a rapidly flashing light is introduced through the telescope. The light is triggered by a stethoscope placed on the patient's larynx that records the vibration of the vocal folds. This technique allows the rapid vibration of the mucosa along the free edge of the vocal cord to be seen in slow motion and facilitates a much more detailed examination of any movement disorder of the mucosa, which may arise from either

movement from the front to the back of the vocal cord or from the free edge into the laryngeal ventricle. Small mucosal or subepithelial lesions that cannot be identified with constant light can often be identified using endostroboscopy.

Electro-laryngography (laryngeal EMGs) can be used to assess cases of vocal fold paralysis but there are doubts about the sensitivity and reproducibility of this investigation. However, when expertly performed, information about the neuromuscular junctions in an immobile vocal fold can be obtained using this technique. The presence of re-innervation potentials indicates that an immobile vocal fold is due to a neurological disorder and can be used to track whether the disorder is likely to return sooner rather than later. Electro-laryngography is also a useful way to isolate specific muscles for injection of botulinum A in patients with laryngeal dystonias.

Flexible transnasal oesophagoscopy is performed under local anaesthesia and is a useful investigation for examining the cricopharyngeal muscle, oesophageal mucosa and the gastro-oesophageal junction in cases of reflux-induced pharyngolaryngitis.

Diagnostic imaging is rarely used as a first-line investigation in the management of voice disorders but can be used to supplement the clinical examination findings. Imaging is most useful in cases of laryngeal trauma and suspected laryngeal neoplasia.

3. Outline the management of this patient.

Table. Measures to improve vocal hygiene

A period of voice rest, especially for acute laryngitis

Hydration with noncaffeinated drinks, especially during periods of heavy voice use

Limiting of vocal loudness (e.g. use of a microphone for teaching or public speaking)

Avoidance of habitual and frequent throat clearing

Avoidance of irritants, including passive smoking

Humidified environment (e.g. use of menthol or eucalyptus inhalations)

Avoidance of medications that have an anticholinergic drying effect

Not smoking

Speech therapy or vocal training, especially for professional voice users

The majority (almost 90%) of patients improve on vocal hygiene measures alone but some may need to go on to surgical excision of the vocal cord nodules.

CASE SIX

Short case number: 4_27_06

Category: Respiratory and ENT systems

Discipline: Surgery_ENT

Setting: ENT department of teaching hospital

Topic: Thyroid nodule, surgical complications [SDL]

CASE

Beverly is a 55 year old nurse who has been followed up for several years because of her thyroid nodules. These run in the family, with her daughter being the latest to join the "T-nodules club".



All her previous investigations including thyroid function had been normal. She is presenting now because, over the last few months, she has noticed a lump in her neck which is starting to embarrass her. More recently she has noted pressure over the lower neck when lying down and she has occasional difficulty swallowing. Her voice is not affected and is otherwise well.

On examination you find a prominent thyroid nodule over the midline and also smaller ones in the right and left lobes. There were no palpable lymph nodes.

QUESTIONS

1. Most patients with thyroid nodules do not have any significant symptoms. Among those with symptoms what are the commonest presentations? What features in the history would raise suspicion of a malignancy?
2. What types of investigations are commonly used in the differential diagnosis of thyroid nodules?
3. Describe the follow up of thyroid nodules. What are the indications for thyroidectomy? Outline the main complications following thyroid surgery.
4. Beverly is worried about developing thyroid cancer and asks if she is at risk. What information would you give her?

Resources:

1. Delbridge L. How to treat: Thyroid nodules. Australian Doctor, September 2009, pp 25-32.
2. Brennan M and French J. Thyroid lumps and bumps. Australian Family Physician. July 2007 Vol 36(7): 531-6.
3. Bova R and Weissberger A. Thyroid cancer an update. Medicine Today, September 2010, Vol 11(9): 27-38.

1. Most patients with thyroid nodules do not have any significant symptoms. Among those with symptoms what are the commonest presentations? What features in the history would raise suspicion of a malignancy?

THYROID nodules are a very common finding, being found on clinical examination in about 7% of the adult population, or in more than 50% of the population aged over 50 if ultrasound is used as a screening modality. The prevalence of thyroid nodules increases with age and is generally higher in women than in men.

Most patients with thyroid nodules do not have any significant symptoms. The absence of symptoms however, does not exclude malignancy. Symptoms that are significant if present include:

- noticeable swelling
- feeling of pressure in the neck
- obstruction such as difficulty swallowing (may indicate impingement of the oesophagus)
- shortness of breath (may indicate impingement of the trachea).

2. What types of investigations are commonly used in the differential diagnosis of thyroid nodules?

Following history and examination, the main investigations in a patient with a thyroid nodule are:

- Thyroid ultrasound
- Serum TSH levels
- Fine needle aspiration of dominant or suspicious nodules
- Radionuclide scan indicated only if the patient is hyperthyroid (suppressed TSH). Routine use in euthyroid patients should be avoided.

Ultrasound - The role of ultrasound in the investigation of the clinically abnormal thyroid gland is increasing and it is now recommended as part of the routine investigation of a palpable thyroid nodule. Ultrasound allows correlation of the clinical features with the ultrasound appearance, allows the nodule to be characterised further, and may detect other significant nodules that are not palpable. The risk of malignancy is the same for nodules found as incidental impalpable lesions as for nodules of the same size that are palpable. Therefore it is important to assess each nodule individually on ultrasound, looking for features suspicious of malignancy. While there is no single ultrasound feature that reliably predicts malignancy, ultrasound features that increase the risk of a nodule being malignant include:

- irregular margins
- presence of microcalcification
- hypo-echogenicity
- absence of a halo
- predominantly solid composition
- intranodule vascularity.¹⁰

As ultrasound alone is limited in its ability to distinguish benign from malignant nodules, further investigation of palpable nodules with FNAB is usually required even when the lesion has a benign appearance on ultrasound.

Fine needle aspiration biopsy (FNAB) is indicated for nearly all palpable and symptomatic nodules and should be considered in other nodules >1 cm. Fine needle aspiration biopsy should also be performed for smaller nodules (<1 cm) that have suspicious clinical or ultrasound features. The management of multiple incidental thyroid nodules seen on ultrasound is controversial. Biopsy of all these nodules is neither practical nor necessary. Focus should be on nodules that show concerning features on ultrasound and other nodules should probably be monitored with ultrasound. Fine needle aspiration biopsy is a simple and useful test but its usefulness is dependent on obtaining an adequate specimen and having it examined by an experienced cytopathologist. The accuracy of FNAB increases when it is performed under ultrasound rather than manual guidance. Fine needle aspiration biopsy may be seen as an accurate test to determine which nodules should be managed surgically and which may safely be observed. Nodules that are benign on FNAB and show no suspicious features on clinical and ultrasound assessment may be observed. Those showing atypical or malignant cytology should be surgically removed.

**3. Describe the follow up of thyroid nodules. What are the indications for thyroidectomy?
Outline the main complications following thyroid surgery.**

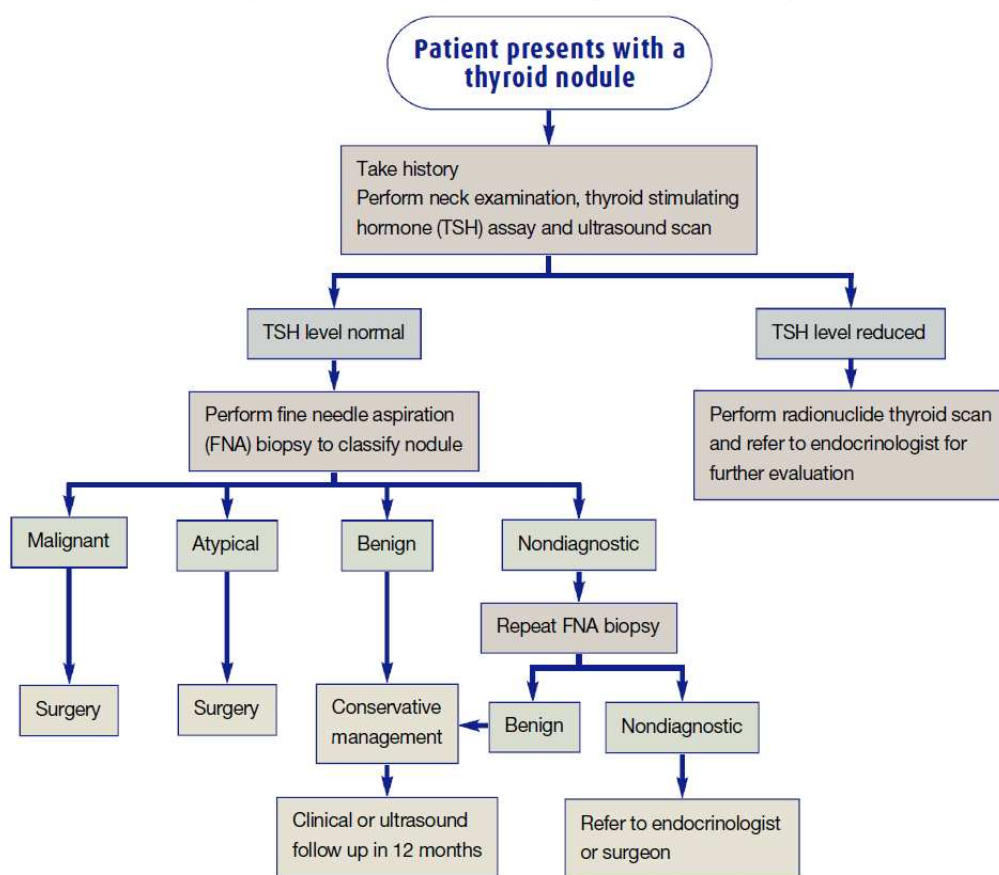
Thyroid nodules with a benign appearance on ultrasound and a benign cytology result on FNAB are usually observed. There is no evidence on the optimal interval and method of follow up. Surveillance generally includes repeat assessment with clinical examination and ultrasound after a period of time. Some specialists do this review at 6–12 months and others recommend review 2–3 years later. If the lesion has increased in size on review, repeat FNAB is recommended. Increase in size, however, is not always a sinister sign as many (89% in one study) cytologically benign nodules increase in size slowly over time. There is an increasing trend to repeat the FNAB at follow up examination even for lesions that are stable on review. Nodules that are benign again on repeat FNAB may be dismissed from further specific follow up and specialist review unless they become symptomatic. Lesions that are stable on follow up ultrasound but are not assessed with repeat FNAB are usually observed for longer before being dismissed from follow up.

Surgery is indicated when there is/are:

- malignancy proven on FNAB
- possible malignancy (atypical, suspicious findings on FNAB)
- local pressure symptoms of compression from the nodule (dysphagia/tracheal compression, persistent cough or choking sensation, pressure on the great veins of the neck, sensation of fullness particularly with arms raised)
- hyperthyroidism (this may be treated with surgery or with radioactive iodine or medications)
- patient choice (usually for cosmetic reasons when nodules are >3 cm)
- other indications (eg. when an adequate sample cannot be obtained by FNAB on several attempts or when a nodule that has shown benign features on ultrasound and FNAB has grown during a period of observation).

If the pressure is due to a large single colloid nodule, hemithyroidectomy is appropriate, whereas if the nodule is a dominant nodule in a multinodular goitre, an initial total thyroidectomy is preferable. A confident diagnosis of malignancy mandates a total thyroidectomy, with or without selective lymph node dissection.

Investigation and management of a patient with a thyroid nodule



Thyroid surgery requires meticulous care to avoid damage to surrounding structures but is now a low risk procedure in experienced hands. Complications include:

- haemorrhage (an uncommon but potentially fatal complication – occurs in <1% of cases)
- voice disturbance or hoarseness due to injury to the recurrent laryngeal nerve (may be temporary or permanent – permanent damage occurs in <1% of cases). Injury to the external branch of the superior laryngeal nerve can cause less obvious voice changes. Patients may complain of an inability to project their voice and easy fatigability of their voice
- hypocalcaemia due to damage to the parathyroid glands or their blood supply (requires calcium supplementation; may be temporary or permanent [2–3%])
- hypothyroidism is expected and is permanent following total thyroidectomy. It may also occur temporarily or permanently following hemithyroidectomy.

4. Beverly is worried about developing thyroid cancer and asks if she is at risk. What information would you give her?

The following features of the history increase the suspicion of the nodule being malignant:

- extremes of age (<20 or >70 years)
- male gender
- family history of: thyroid malignancy or genetic disorders such as syndrome of multiple endocrine neoplasia type 2, familial polyposis coli and Cowden syndrome
- prior exposure to radiation, including radiotherapy to the head and neck (eg. previous treatment of Hodgkin disease affecting the neck).

In a patient with a thyroid nodule, certain clinical findings which would indicate a higher likelihood of malignancy:

- Large nodules >4 cm
- Firm or hard nodules, especially if fixed to adjacent neck structures
- Hoarseness due to vocal cord paralysis
- Rapidly growing nodules (may suggest anaplastic carcinoma or lymphoma)
- Nodules associated with enlarged cervical lymph nodes

Ultrasound findings that are suspicious for malignancy include:

- Nodules with irregular borders
- Central hypervascularity
- Microcalcification within the nodule

Key aspects of thyroid cancer treatment and follow up

- Surgery is the mainstay of treatment and total thyroidectomy is indicated in most patients with papillary or follicular cancer of the thyroid. Experienced thyroid surgeons should perform the surgery.
- Radioactive iodine ablation and long-term thyroid stimulating hormone (TSH) suppression using thyroxine are useful adjuvant therapies. GPs need to be aware of the target TSH level determined by the endocrinologist to avoid inadvertently reducing the thyroxine dose because of the mistaken interpretation that the TSH is too low.
- The increasing use of recombinant TSH (rhTSH) is obviating the need for patients having to stop taking thyroxine and being rendered temporarily hypothyroid so they can have stimulated thyroglobulin (Tg) levels measured and/or have follow-up whole body radioisotope thyroid scans to monitor for residual or recurrent thyroid cancer. Use of rhTSH also avoids the patient being hypothyroid between thyroidectomy and radioactive iodine ablation.
- External beam radiotherapy and chemotherapy are rarely indicated.